

ZUERICH-FLUNTERN, Switzerland

WMO: 066600

Lat: 47.3781N

Lon: 8.5658E

Elev: 558

StdP: 94.80

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.1	-6.2	-11.0	1.6	-6.7	-9.2	1.8	-4.8	12.0	7.9	10.6	8.1	2.2	50	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	29.8	19.5	27.8	18.8	26.0	18.1	20.3	27.4	19.5	26.0	18.8	24.5	2.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.9	22.8	17.3	13.2	21.8	16.7	12.7	21.2	60.8	27.6	58.0	26.2	55.5	24.8	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	6.9	5.5	DB	-10.3	32.4	2.7	1.7	-12.3	33.6	-13.9	34.6	-15.5	35.6	-17.5	36.8	(4)
(5)			WB	-10.8	22.0	2.7	1.0	-12.7	22.7	-14.3	23.3	-15.8	23.8	-17.7	24.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	1.0	2.1	6.1	9.9	14.0	17.7	19.3	19.0	14.8	10.7	5.2	1.8	(6)	
	DBStd	7.59	4.10	4.39	4.15	4.16	3.75	3.95	3.39	3.40	3.35	3.72	3.78	3.89	(7)	
	HDD10.0	1137	279	222	131	53	8	1	0	0	3	37	148	254	(8)	
	HDD18.3	3144	537	455	379	253	142	59	29	33	114	237	393	512	(9)	
	CDD10.0	1203	1	1	11	51	132	231	289	278	147	59	5	0	(10)	
	CDD18.3	168	0	0	0	0	7	40	60	53	8	0	0	0	(11)	
	CDH23.3	1151	0	0	0	5	58	293	416	346	34	0	0	0	(12)	
	CDH26.7	293	0	0	0	0	6	77	113	96	1	0	0	0	(13)	
(14) Wind	WSAvg	2.0	2.3	2.3	2.5	2.1	2.0	1.8	1.8	1.6	1.7	1.7	2.0	2.3	(14)	
(15) Precipitation	PrecAvg	1120	66	66	72	87	116	128	125	130	92	77	82	81	(15)	
	PrecMax	1549	156	289	237	200	287	232	285	295	229	188	211	169	(16)	
	PrecMin	860	0	10	13	6	18	34	37	27	23	7	0	1	(17)	
	PrecStd	158	39	46	44	46	49	50	54	56	48	44	46	45	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	13.7	18.9	23.9	27.3	31.5	32.2	32.3	26.2	21.3	16.0	11.7	(19)	
		MCWB	8.2	7.7	10.5	13.4	17.0	20.0	20.0	19.9	18.0	15.0	11.1	8.0	(20)	
	2%	DB	10.0	11.5	16.5	21.4	25.0	28.9	29.6	29.2	24.3	19.0	13.6	10.0	(21)	
		MCWB	6.9	6.8	9.4	12.3	16.1	19.0	19.5	19.6	17.4	14.3	10.0	7.5	(22)	
	5%	DB	8.1	9.7	14.3	19.2	22.8	26.7	27.5	27.1	21.9	17.2	11.7	8.6	(23)	
		MCWB	5.6	6.0	8.7	11.4	14.7	18.3	18.8	19.0	16.2	13.5	9.2	6.6	(24)	
	10%	DB	6.5	7.9	12.3	17.0	20.7	24.6	25.6	24.9	20.0	15.5	10.1	7.2	(25)	
		MCWB	4.6	5.0	7.7	10.4	13.9	17.3	18.0	18.2	15.6	12.8	8.2	5.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	8.7	11.5	14.3	18.0	21.5	21.5	21.3	19.1	16.1	11.7	9.1	(27)	
		MCDB	11.5	12.4	16.9	22.2	25.6	29.5	29.5	28.4	24.7	19.4	14.6	10.5	(28)	
	2%	WB	7.2	7.2	10.2	12.8	16.7	19.8	20.2	20.3	17.9	15.0	10.5	7.8	(29)	
		MCDB	9.3	10.7	14.8	19.6	23.2	26.5	27.3	27.3	22.6	17.9	12.8	9.5	(30)	
	5%	WB	5.8	6.2	9.1	11.9	15.6	18.9	19.4	19.5	16.9	14.0	9.4	6.7	(31)	
		MCDB	7.8	9.2	13.4	18.0	21.1	25.1	25.8	25.7	20.8	16.6	11.4	8.4	(32)	
	10%	WB	4.7	5.2	8.1	10.9	14.5	18.0	18.6	18.6	16.0	13.1	8.4	5.6	(33)	
		MCDB	6.4	7.7	11.8	16.1	19.4	23.6	24.3	23.8	19.3	15.0	10.0	7.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	5.7	7.5	8.8	8.8	9.1	9.0	8.6	7.6	6.1	4.6	4.0	(35)	
		MCDBR	6.2	8.8	10.9	12.6	12.3	12.4	12.2	12.0	10.4	8.4	7.1	5.6	(36)	
	5% WB	MCWBR	4.1	5.1	5.6	5.8	5.5	5.0	4.6	4.5	4.7	4.4	4.3	3.8	(37)	
		MCWBR	5.6	7.7	9.6	11.5	11.0	11.3	11.1	10.9	9.3	7.7	6.5	5.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.287	0.311	0.346	0.380	0.394	0.399	0.401	0.393	0.368	0.342	0.311	0.283	(40)	
		taud	2.445	2.381	2.347	2.291	2.296	2.322	2.309	2.346	2.400	2.459	2.492	2.456	(41)	
	Ebn at Noon		807	854	872	872	870	866	857	849	839	808	769	766	(42)	
		Edh at Noon	66	86	105	122	126	124	124	114	99	80	63	58	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.19	2.04	3.25	4.34	4.97	5.72	5.54	4.83	3.63	2.29	1.29	0.98	(44)	
	RadStd		0.13	0.27	0.44	0.63	0.50	0.53	0.50	0.49	0.37	0.25	0.16	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.42	N/A	N/A	N/A	N/A	N/A
(47) Regional (119 neighbors)		N/A	N/A	N/A	+0.71	+0.42	+0.34	N/A	N/A	+57	+16

Nomenclature: See separate page